Engineering Economics

Lecture 1 — Terminologies, Principles & Decision-Making

1. WHAT IS ECONOMICS?

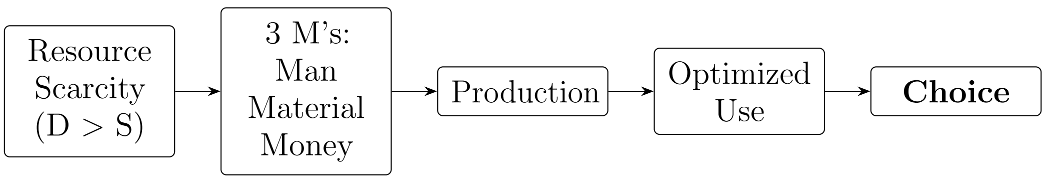
> **Hook:** Every engineering project consumes **Man, Material & Money (3 M's)**. These are *scarce* (Demand > Supply), so engineers must make **optimized choices**. That is exactly what Economics studies.

1.1. Classical Definitions

Economist	Definition	Year
Adam Smith	"Science of Wealth"	1776
A. Marshall	"Science of Welfare"	1890
L. Robbins	"Science of Scarcity & Choice"	1930

Note: Definitions evolve from Wealth → Welfare → Scarcity & Choice. Robbins's (1930) is the **modern/neo-classical** definition.

1.2. From Scarcity to Choice — Conceptual Chain



2. MICRO & MACRO ECONOMICS

Micro Economics	Macro Economics
Individual/firm level	Country/institution level
Classical view (pre-1930)	Keynesian view (post-1930)
"Rich people ⇒ Rich country"	"Rich country ⇒ Rich people"
No governance needed	Governance is a necessity

2.1. The 1930 Turning Point — Great Depression

Micro (Classical) Individual focus $\xrightarrow{\text{market fails}}$ **1930** Great Depression Deadlock $\xrightarrow{\text{Keynes}}$ **Macro** (Keynes) Aggregate focus

Key Point: J.M. Keynes shifted economics from Micro to Macro after 1930. Critical decisions *must* be made by government for **common welfare** (governance necessity).

3. ENGINEERING ECONOMICS

> **Definition:** Engineering Economics **measures the cost and benefit of an engineering project** to support rational decision-making.

Decision Rule:

$$\text{Benefit} > \text{Cost} \implies \textbf{Accept} \qquad \text{Benefit} < \text{Cost} \implies \textbf{Reject}$$

4. ROLE OF ENGINEERING ECONOMICS IN DECISION-MAKING

1. **Measure Cost & Benefit** of an engineering project.
2. **Select Project within Budget Limitations.**
3. **Select Lease or Purchase Option.**
   Short-term need ⇒ Lease — Long-term need ⇒ Purchase
4. **Know Project Risks** — quantify and plan for uncertainties.
5. **Decide Product Expansion or Not** — requires **demand forecasting**.

Key Point: More competition ⇒ lower price: market dynamics must be studied.

5. SEVEN PRINCIPLES OF ENGINEERING ECONOMICS

*These are guidelines for **how** to think through a decision — not what to decide. Use as a checklist on every project.*

1. **Develop the Alternatives**
   List all candidate projects: X, Y, Z ...

2. **Focus on Differences**
   Compare only what differs between alternatives.

3. **Use Consistent Viewpoints**
   One lens only: profit/loss or benefit/cost.

4. **Use a Common Unit of Measurement**
   Currency or quantity — never mix.

5. **Consider All Relevant Criteria**
   Environmental & social values (EIA validation).

6. **Make Uncertainty Explicit**
   State assumptions; X selected → Implement → Result.

7. **Revisit Your Decision**
   Self & Post Evaluation → for better future projects.

QUICK REFERENCE TABLE

Topic	Key Takeaway
Classical Definitions	Wealth (1776) $\rightarrow$ Welfare (1890) $\rightarrow$ Scarcity & Choice (1930)
Scarcity	D > S $\Rightarrow$ scarce resources $\Rightarrow$ need for optimized choice
Micro vs Macro	Individual/firm vs aggregate/national economic behaviour; Many Micros $\Rightarrow$ Macro
Keynes (1930)	Market failed $\Rightarrow$ government must act for common welfare
Eng. Economics	Measures cost & benefit to support rational project decisions
Decision Rule	Benefit > Cost: Accept | Benefit < Cost: Reject
7 Principles	Systematic checklist: Alternatives $\rightarrow$ Differences $\rightarrow \cdots \rightarrow$ Revisit
5 Roles	Measure, Budget, Lease/Buy, Risk, Demand Forecast

"Economics is the science of making decisions in the presence of scarce resources."
— Robert Dorfman